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Started	Friday, 9 October 2026, 9:15 AM
Completed	Friday, 9 October 2026, 9:19 AM
Duration	3 mins 48 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1
b. log n
c. n/2
d. n ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search
b. Linear search
c. Binary search ✓
d. Sequential search

Question **3**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is small and unsorted ✓
c. When the list is small and sorted
d. When the list is large and unsorted

Question **4**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist
b. The search continues in the right sublist
c. The search ends successfully ✓
d. The list is sorted

Question **5**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By comparing each element in the list sequentially ✓
c. By dividing the list into halves
d. By sorting the list first

Question **6**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search has a lower time complexity on large, sorted lists ✓
c. Binary search can find multiple instances of the target element
d. Binary search works on unsorted lists

Question **7**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O👎
b. O(1)
c. O(n log n)
d. O(log n) ✓

Question **8**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

- a. Searching ✓
- b. Discovering
- c. Finding
- d. Mining

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. There must be a mechanism to access middle element directly
- b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- c. Must use a sorted array
- d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **10**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 500
- b. 100
- c. 1000
- d. 10 ✓

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Sorting the list before searching
- b. Dividing the list in half repeatedly
- c. Skipping every second element
- d. Checking each element sequentially ✓

Question **12**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search stops immediately ✓
- b. The search backtracks to find duplicate elements
- c. The search starts over from the beginning
- d. The search continues until the end of the list

Question **13**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

- a. Bubble ✓
- b. Selection
- c. Complexity
- d. Insertion

Question **14**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Sorting elements in a list
- b. Deleting elements from a list
- c. Determining whether an element is present in a list ✓
- d. Inserting elements into a list

Question **15**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. 1
- b. log n
- c. n ✓
- d. n/2